Transport Project:
Report

Group 5

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1. Introduction and Objectives

This study investigates the diffusion of food dye in water and gel mediums using basic household supplies and devices. Diffusion can be defined as a fundamental process of mass transport and is influenced by a large variety of material properties such as viscosity, structure, and molecular interactions. Conducting and understanding this experiment and the basic principles of diffusion are valuable tools in understanding many fundamental engineering processes and concepts.

This project consists of various aspects which lend insight into mass transport fundamentals. The first objective consisted of experimental design and setup based on the materials list found in the project description. Naturally, the next step was to run the experiments using video technology to capture the radial spread of food dye in the petri dish during each trial. Following experimentation, a theoretical model was developed using appropriate natural phenomena and assumptions for our experimental setup. After completion of the experiments, a numerical simulation was constructed to model the process and calculate the diffusion coefficients. These numerical simulations were compared to the experimental and analytical results, and discrepancies were discussed along with theories for these errors and recommendations for possible improvements in future experiments.

2. Experimental Methods and Observations

Experiments were conducted using a standard Petri dish with a diameter of 60mm and a depth of 15mm. A GoPro camera was used to capture the radial diffusion from directly overhead. A micropipette was used to deliver controlled volumes of food dye into the center of the Petri dish.

A similar procedure was followed for all trials conducted in the experiment. To begin the trial, a droplet of dye was placed in the center of the Petri dish and filming with the GoPro began. Snapshots were taken every 15 seconds in the water trials and every 10 minutes in the gel trials. At these given intervals, the radius of the dye region was measured along with the RGB values from the ImageJ software. Each experimental medium was repeated 3 times to determine the experimental repeatability.

During the trials, a few observations were found. The diffusion of dye in water was much faster than the diffusion of dye in gel. However, the water trials exhibited higher variability between trials, leading to lower repeatability. Due to this, the water trials had to be redone multiple times to account for this lower repeatability. The major improvement to avoid high convection influences was to make the water depth 1/3 of the volume of the petri dish and lowering the amount of dye dropped. This improved the symmetry of the initial drop and consequently the uniform spreading of the dye.

3. Theoretical Formulation and Assumptions

The theoretical modeling for this experiment began with the generalized mass-transport equation. Several assumptions were made based on the experimental design: no reaction occurs; an axis of symmetry exists along the θ-direction; and there is no bulk flow. A fourth assumption—verified by constructing a 3D model—was that the radius is much larger than the height. As a result, the influence of the vertical dimension on diffusion is negligible and can be ignored.

The boundary conditions were defined as follows: zero flux at the outer edge of the Petri dish and a maximum concentration at the center (radius = 0). The initial condition specified that the concentration throughout the domain, except within the initial dye droplet, was zero at the start of the experiment. Additionally, the concentration at the location of the drop was treated as an unlimited source (Equation 3.1).

$$\frac{\partial C_A}{\partial t} = \frac{D_{AB}}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_A}{\partial r}\right)$$

$$BC\ 1\ \left.\left(\frac{\partial C_A}{\partial r}\right)\right]_0 = 0\ \&\ BC\ 2\ \left.\left(\frac{\partial C_A}{\partial r}\right)\right]_{Rmax} = 0\ \&\ IC\ C_A\left(r < R_{Drop},\ 0\right)\ =\ C_{A0}$$

Equation 3.1 Governing Equation, Boundary Conditions, and Initial Condition

Next, the governing equation, boundary conditions, and initial condition were non-dimensionalized. Separation of variables was then applied to decompose the PDE into two ordinary differential equations: a first-order ODE in time and a second ODE in the radial coordinate, which corresponds to a Bessel equation. Because the radial ODE has a known analytical form, it was evaluated in MATLAB using the built-in Bessel functions. This process led to the derivation of the final expression for the concentration field.

$$C(r,t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)\exp(-\alpha\lambda_n^2\tau)\ \ where\ \alpha = \frac{D_{AB}t_0}{R^2}$$

Equation 3.2 Analytical Solution to the PDE

In the numerical simulation, the nondimensionalized equation and boundary conditions are discretized using the finite central difference approximation. Due to the discretization process, the first boundary condition turns into the ghost point boundary condition. Additionally, the implicit Euler method was implemented to calculate the evolution of the system regarding time.

Because of experimental constraints, the initial condition was changed from a fixed amount of food dye to an unlimited-source condition. This adjustment better represented the behavior of the actual system.

4. Data analysis and parameter estimation

To analyze the videos of both the gel and water trials, still frames were extracted at fixed intervals to ensure sufficient and consistent time steps. For the water trials, frames were taken every 15 seconds from 0 to 90 seconds, while the gel trials used 10-minute intervals from 0 to 120 minutes. Depending on the analysis method used (ImageJ or MATLAB), additional processing steps such as cropping and generating masks were required.

The next major step was to create a digital ruler by establishing a ratio between the real-life diameter of the Petri dish (60 mm) and the number of pixels in the dish occupied in each still frame (typically 370–390 px). This ratio was then used to calculate the dye-front radius. Once the radii were determined, a graph illustrating the relationship between radius and time for both the water and gel trials was generated. All trials for each medium were compared, and linear best-fit models were applied, yielding R^2 values of 0.892 for the gel trials and 0.765 for the water trials.

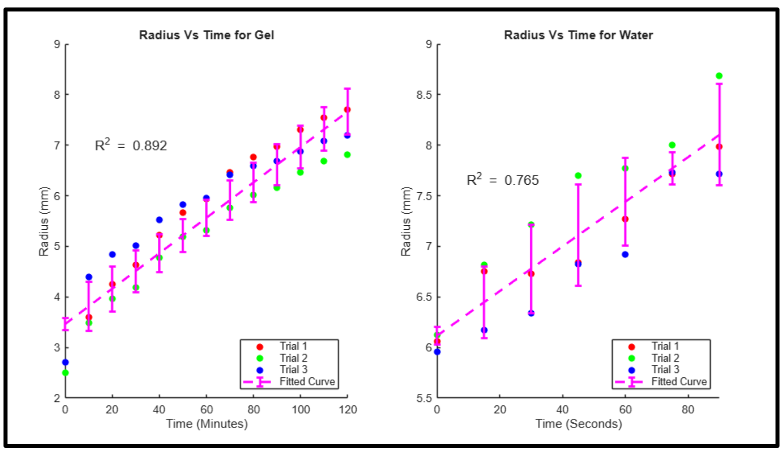

Fig. 1 Radius Vs Time shows the correlation between Radii in mm and time is both minutes and seconds respectively for Gel and Water trials. The Gel graph has an R^2 value of 0.892 with standard deviations ranging from 0.086-0.499. The Water graph has an R^2 value of 0.765 with standard deviations ranging from 0.1194-0.4903.

The dye-front intensity was then normalized by subtracting the RGB value of the clear-water region measured earlier and dividing the result by the maximum pixel intensity in the image (255 for 8-bit images). This produced a nondimensionalized intensity, which can be correlated to the concentration of food dye via Beer-Lambert's Law.

Afterward, the analytical solution to Equation 3.2 was implemented in MATLAB and compared with the numerical solution. Using the same boundary and initial conditions described earlier, both models, Figure 2, show strong agreement. This confirms that either approach can reliably simulate the system.

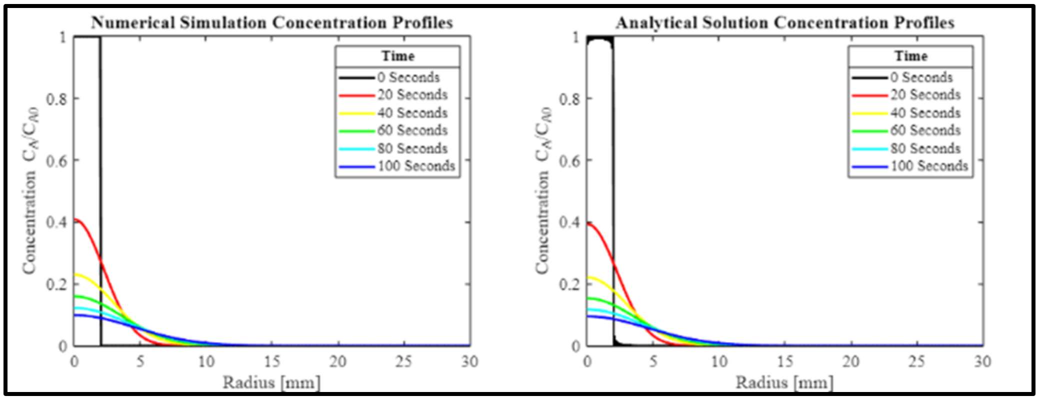

Fig. 2 Comparisons of the numerical and analytical models. Both produced identical results despite using different computational methods. These simulations assume mass conservation in addition to the previously stated boundary conditions.

Because the two models agreed, the numerical simulation was used to estimate the diffusion coefficients. Changing a boundary condition to represent an unlimited source was more straightforward numerically than analytically. The coefficients were then computed using the bisection method. Figure 3 presents the diffusion coefficient for each time-interval and trial. Porosity was subsequently calculated using the trial-specific diffusion coefficients yielding values from 0.407 to 0.6139.

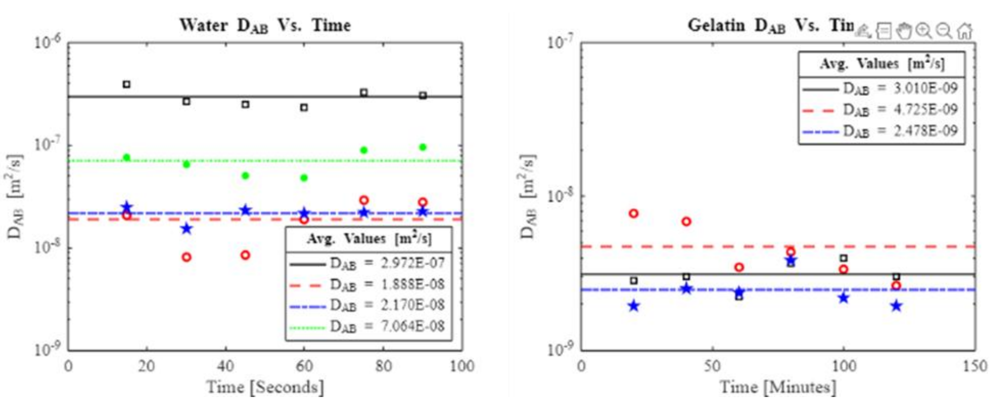

Fig. 3 Shows the calculated diffusion coefficients for the water and gelatin systems. These values were obtained from the measured radii, intensity, and time at each image snapshot. They were then plotted versus time to assess whether time acted as a lurking variable.

5. Discussion of Results

Although efforts were made to minimize errors during the experimental setup, image processing, and data analysis, several limitations remain affected the validity of the models. Diffusion in water was sensitive to the height, angle, and release speed of the dye drop; higher or faster releases introduced convective effects. In addition, the experiment assumed a uniform temperature, which may not hold in practice. Any temperature variation would alter the diffusion coefficient and influence the results.

In processing the data, the diffusion in water was assumed to remain perfectly circular across all experiments. Additionally, the RGB values obtained from ImageJ were highly sensitive to image quality and to the exact cursor position during measurement.

Another source of error arose in the numerical simulation. Because the initial condition was changed to an unlimited-source assumption, mass conservation could not be enforced. As a result, the numerical approximation both loses and gains mass across time steps, and this error was not quantified. This modification was necessary because the image frames showed no change in RGB values within the central drop region, indicating that the dye remained saturated. Consequently, the initial concentration could not be determined.

Medium	**Experiment #**	**Diffusion Coefficient (m^2/s)**
Water	Trial 1	(2.972 ± 0.604)E-07
	Trial 2	(2.170 ± 0.336)E-08
	Trial 3	(1.888 ± 0.909)E-08
	Trial 4	(7.064 ± 1.979)E-08
Gelatin	Trial 1	(3.130 ± 0.630)E-09
	Trial 2	(4.725 ± 2.070)E-09
	Trial 3	(2.478 ± 0.725)E-09

Table 1 Calculated diffusion coefficients in water and gelatin per trial.

The average diffusion coefficient in water was calculated as 1.021 x 10^{-7}, which falls outside the literature range of 4.4x10^{-8} to 1.7x10^{-10} m^2/s. This is likely due to the inaccuracies as stated beforehand. For gelatin, the average diffusion coefficient was 3.44x10^{-9} m^2/s which is consistent with the reported value of 2x10^{-9} m^2/s for agar gels.

Another parameter that was calculated was the porosity of the gelatin systems. This was accomplished with equation 3.3.

$$\varphi = \sqrt[2]{\frac{D_{AB,Gel}}{D_{AB,Water}}}$$

Equation 3.3 Porosity calculation via diffusion coefficients

These results overestimate the true porosity because the water diffusion coefficients were roughly three times higher than expected from literature (0.213). In addition to the diffusion equation not fully representing the physical system, variability in the gel preparation, especially the ratio of water to gel powder, may also contribute to error.

Lastly, a separate numerical model was developed to assess the influence of height on dye diffusion in both the water and gelatin systems (section 7.3). The model showed that, in water, the dye diffuses only radially in both short- and long-term simulations. In gelatin, the dye initially exhibits spherical diffusion, which later transitions to predominantly radial diffusion. These results support the earlier assumption that the effect of height on diffusion is negligible. More information about the 3D model can be found in the appendix.

6. Conclusions and recommendations

This project quantified dye diffusion in water and gel using image analysis and numerical modeling. Through our experimental approach and image processing, we concluded that diffusion through the water medium was 30 times larger than in gel with an average value of 1.021×10^{-7} for water and 3.44×10^{-9} m^2/s for gelatin. This difference arises from the restricted mobility in gelatin.

We also observed that convective forces were difficult to eliminate for water trials. Small inconsistencies in the angle drop, release height, and dropping speed introduced flow patterns that influenced early diffusion behavior. This is backed up by the R^2 value of 0.765 in figure 1 for the water trials.

Additionally, our method of determining the concentration through the analysis of RGB values introduced an upper limit to the application of Beer-Lambert's Law. At high dye concentrations, the center region of the drop remained saturated throughout the experiments, thus preventing the accurate calculation of initial concentration, and complicating the development of an exact analytical concentration profile. Nevertheless, by nondimensionalizing concentration, we were able to compute a theoretical profile using both a Bessel function approximation and our numerical model. These two results were nearly identical, demonstrating the validity of our numerical approach (figure 2).

While our results successfully demonstrate a clear difference in diffusion of food dye through gel and water mediums, it also reveals the sensitivity of the system in real-world experimental conditions. Future trials could use a more consistent droplet-release method and improved lighting for better image processing. Additionally, future work could explore other aspects of diffusion, such as how dye color influences diffusion rates, or how different gel concentrations modify transport behavior.

References

Welty, J. R.; Rorrer, G.; Foster, D. Fundamentals of Momentum, Heat, and Mass Transfer, 7th ed.; Wiley: Hoboken, NJ, 2023.

ImageJ, version 1.54; National Institutes of Health: Bethesda, MD.

MATLAB, R2025a; The MathWorks, Inc.: Natick, MA, 2025.

[1]https://www.researchgate.net/publication/222236497_Diffusion_coefficients_for_direct_dyes_in_aqueous_and_polar_aprotic_solvents_by_the_NMR_pulsed-field_gradient_technique

[2] https://digitalcommons.andrews.edu/cgi/viewcontent.cgi?article=1378&context=autlc

Appendix

3-Dimension Model (Numerical Simulation)

A three-dimensional numerical simulation was constructed to evaluate how the dish height influences diffusion. The model included time, radial position, and vertical height. The first case examined diffusion over a long period. For water, the dye settled at the bottom and spanned the full height; for gelatin, the dye remained on the surface. Figure 4 shows that, in both cases, the concentration within the diffusion front varies only radially and not with height.

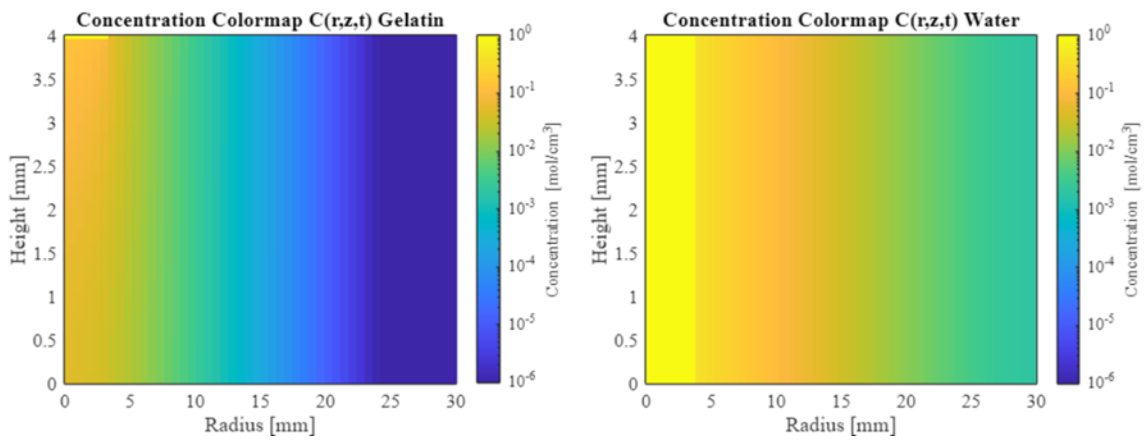

Fig. 4 Diffusion front for both the water and gelatin systems after long diffusion times.

The second case examined the initial stage of diffusion. In water, the dye immediately occupies the entire vertical height, consistent with experimental observations, yet diffusion again proceeds only radially. In gelatin, the dye exhibits spherical diffusion. These results indicate that height does not significantly influence dye transport within the system.

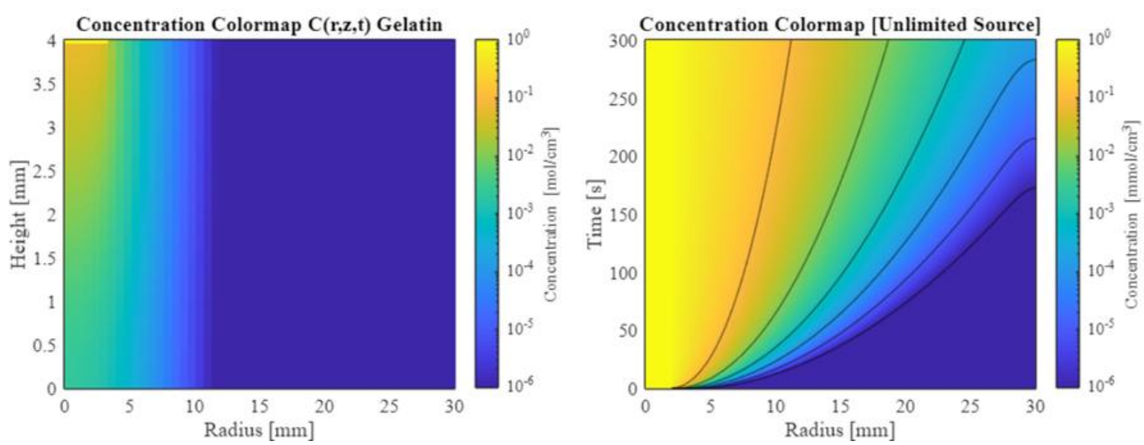

Fig. 5 Left: spherical diffusion in the gelatin system at a short time interval. Right: diffusion front in the water system under an unlimited-source condition.

Therefore, the experiment can be accurately represented using the 2D model. This approach simplifies the calculations while still capturing the essential physics of the system.

2-Dimensional Numerical Simulation

```
clc; clear all; close all; format short; %#ok<CLALL>
```

Variable Declination

```
% Constants
DAB = 0.1; % Diffusion constant (cm^2/s)
t0 = 500; % Duration of experiment (S)
R = 30; % Radius of petry dish (mm)
CA0 = 1; % Concentration of the Food coloring (mole/cm^3)
R_Drop = 1.00; % Radius of food coloring drop (mm)

% Resolution of model
NR = 500; % # of points/resolution in the Non-dimensionalized Radial dimension
NT = 500; % # of points/resolution in the Non-dimensionalized Time dimension

% Dimensions
Gamma = linspace(0,1,NR); % Non-dimensionalized Radial (r/R)
Tau = linspace(0,1,NT); % Non-dimensionalized Time (t/t0)

% Delta variables
dR = 1 ./ (NR - 1); % Delta Phi
dT = 1 ./ (NT - 1); % Delta Tau

% Defining Matricies
Coefficent_Matrix = zeros(NR, NR); % Coefficent Matrix
Concentration = zeros(NR, NT); % Concentration Matrix

% Portion of the dish's radi that is initially occupied by the drop
A = max(floor(NR*R_Drop/R) + 1 , 2);
```

Building the Coefficent Matrix

```
% Defined Constants in the General Equation
Alpha = -DAB*t0*dT/R^2; % Reacuring constant
a = @(i) Alpha .* (1/dR^2 - 1./(2.*Gamma(i).*dR)); % Coefficent for Cn,r-1
b = 1 - 2*Alpha/dR^2; % Coefficent for Cn,r
c = @(i) Alpha .* (1/dR^2 + 1./(2.*Gamma(i).*dR)); % Coefficent for Cn,r+1
d = 2*Alpha/dR^2; % Boundary Conditions Constant at Phi = 0 or Phi = 1 of Cn,r+1 or Cn,r-1

% Building the A matrix
for i = 1:NR

if i == 1; Coefficent_Matrix(i,i:i+1) = [b d]; end % Boundary condition at Phi = 0
if i ~= 1 && i ~= NR; Coefficent_Matrix(i,i-1:i+1) = [a(i) b c(i)]; end % Main Equation
if i == NR; Coefficent_Matrix(end,end-1:end) = [d b]; end % Boundary condition at Phi = 1

end
```

Calculating the Concentration at each Point (Conservation of Mass)

```
% Initial Condition (At Phi = 0, the concentration is always equal to 1. Unlimited source of
food coloring)
Concentration(1:A,1) = 1;

% Converting the Coefficent matrix into a sparse matrix to save memory
Coefficent_Matrix = sparse(Coefficent_Matrix);

% Mass matrix
Mass = zeros(NR,NT);
Mass_0 = Concentration(1,1)*pi()*(A*dR)^2;

% Calculating the concentration at each point for the next time step
for j = 2:NT

Concentration(:,j) = Coefficent_Matrix\Concentration(:,j-1);

% Conservation of Mass check/correction
for k = 1:NR
Mass(k,j) = Mass(k,j) + Concentration(k,j)*pi()*((k*dR)^2 - ((k-1)*dR)^2);
end

Concentration(:,j) = Concentration(:,j).*(Mass_0/sum(Mass(:,j)));

end
```

Calculating the Concentration at each Point (Unlimited Source)

```
% Initial Condition (At Phi = 0, the concentration is always equal to 1. Unlimited source of
food coloring)
Concentration(1:A,1) = 1;

% Converting the Coefficent matrix into a sparse matrix to save memory
Coefficent_Matrix = sparse(Coefficent_Matrix);


% Calculating the concentration at each point for the next time step
for j = 2:NT
Concentration(:,j) = Coefficent_Matrix\Concentration(:,j-1);
Concentration(1:A,j) = 1;

end
% Portion of the dish's radi that is initially occupied by the drop
A_Drop = max(floor(NG*R_Drop/R) + 1 , 1);
```

Building the Coefficent Matrix

```
% Defined Constants in the General Equation
Alpha = -DAB*t0*dT/R^2; % Reacuring constant
```

```
Beta = -DAB*t0*dT/H^2; % Reacuring constant

a = @(i) Alpha .* (1/dG^2 - 1./(2.*Gamma(i).*dG)); % Coefficent for Cn,r-1,z
b = 1 - 2*Alpha/dG^2 - 2*Beta/dP^2; % Coefficent for Cn,r,z
c = @(i) Alpha .* (1/dG^2 + 1./(2.*Gamma(i).*dG)); % Coefficent for Cn,r+1,z
d = Beta/dP^2; % Coefficent for Cn,r,z+1 and Cn,r,z-1
e = 2*Alpha/dG^2; % Boundary Condition constant at Gamma = 0 or 1 of Cn,r+1,z or Cn,r-1,z
f = 2*Beta/dP^2; % Boundary Condition constant at Phi = 0 or 1 of Cn,r,z+1 or Cn,r,z-1


% Building the Coefficent matrix
% Cordinate system of the cylinder.
% r = 1: First point of the radius (center).
% r = NG: last point of the radius (furtherest point from the origion)
% z = 1: Top of the cylinder (Highest Point)
% z = NP: Bottom of the cylinder(Lowest point)

for z = 1:NP
for r = 1:NG

Position = NG*(z - 1) + r; % Position in the matrix

% Corner Conditions
% if z == 1 && r == 1; Coefficent_Matrix(1,1) = [1]; end % Boundary Condition at Gamma = 0
and Phi = 1 (Unlimited NA)
```

Equations used for Image Processing

$$Scale_i = \frac{Diameter_{true}}{Length_{i,\ Pixels}}$$

$$Radius_{pixels} = \sqrt{\frac{\sum mask\ image}{\pi}}$$

$$Radius_i = Radius_{pixels} \times Scale_i$$

$$Radius_{Effective} = \frac{R_x + R_y}{2}$$

$$Average\ Intensity = \frac{R + G + B}{255}$$

$$Intensity_{normalized} = \frac{I_{dye}}{I_{background}}$$